



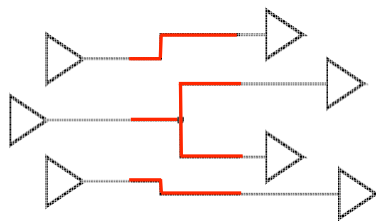
Wires

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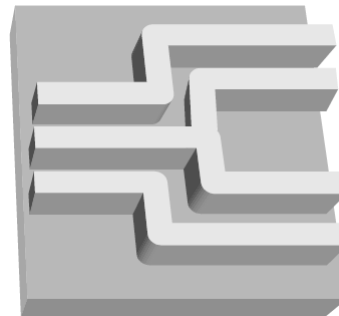
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11

The Wire



Schematic



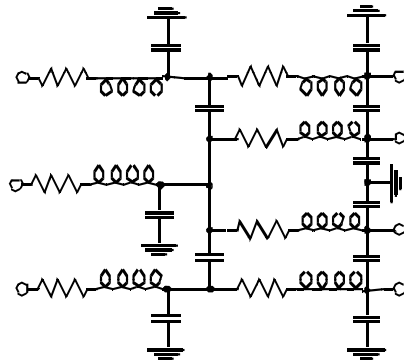
Physical

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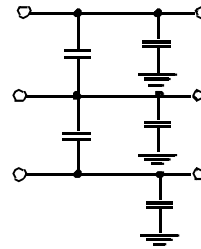
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12

Wire Models



All-inclusive model

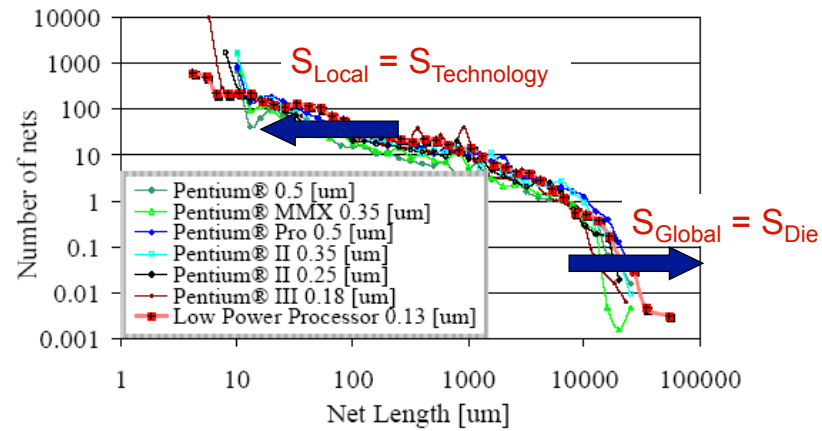


Capacitance-only

Impact of Interconnect Parasitics

- ❑ Interconnect and its parasitics can affect all of the metrics we care about
 - Cost, reliability, performance, power consumption
- ❑ Parasitics associated with interconnect:
 - Capacitance
 - Resistance
 - Inductance

Interconnect Length Distribution



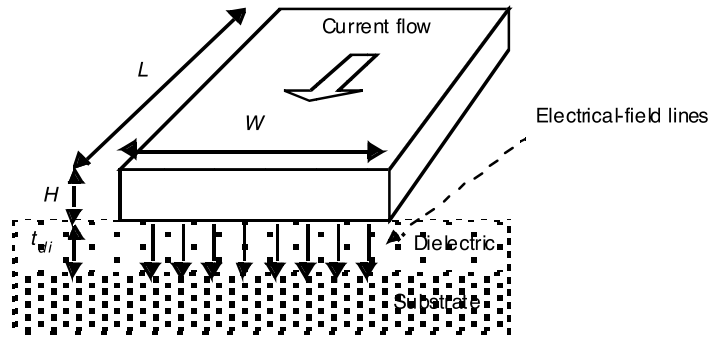
From Magen et al., "Interconnect Power Dissipation in a Microprocessor"

INTERCONNECT

Capacitance



Capacitance: The Parallel Plate Model



$$C_{int} = \frac{\epsilon_{di}}{t_{di}} WL$$

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17

Permittivity

Material	ϵ_r
Free space	1
Aerogels	~ 1.5
Polyimides (organic)	3-4
Silicon dioxide	3.9
Glass-epoxy (PC board)	5
Silicon Nitride (Si_3N_4)	7.5
Alumina (package)	9.5
Silicon	11.7

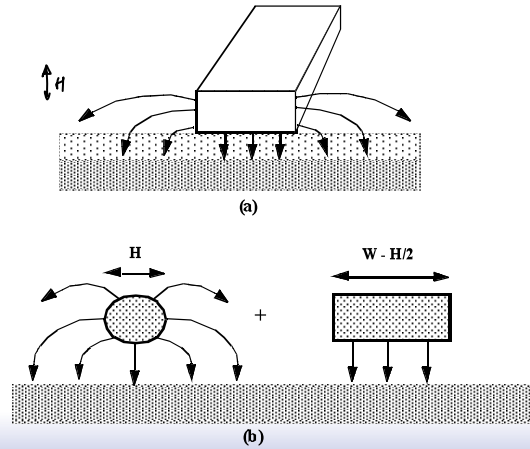
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18

Fringing Capacitance

$$c_{wire} = c_{pp} + c_{fringe} = \frac{w\epsilon_{di}}{t_{di}} + \frac{2\pi\epsilon_{di}}{\log(t_{di}/H)}$$

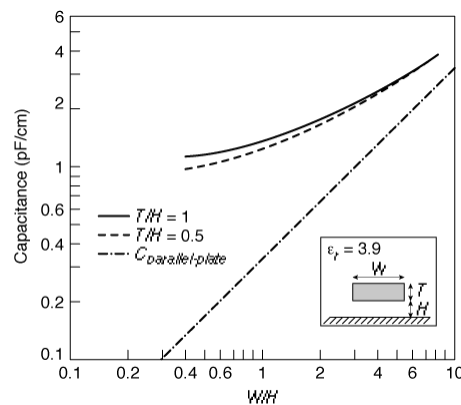


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19

Fringing versus Parallel Plate



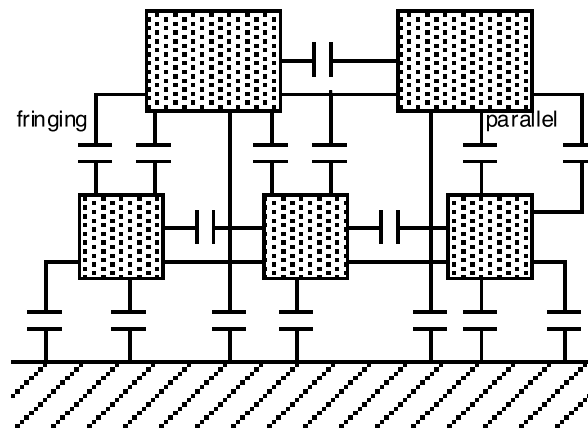
(from [Bakoglu89])

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20

Interwire Capacitance



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21

Capacitive coupling and noise

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22

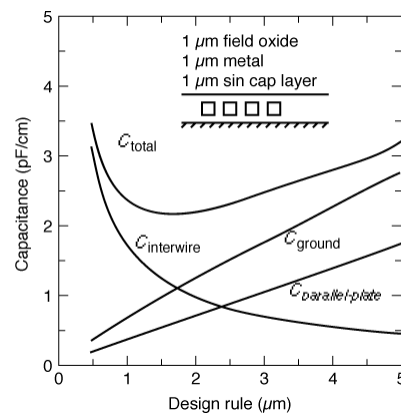
Coupling Capacitance and Delay

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Impact of Interwire Capacitance



(from [Bakoglu89])

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24

Wiring Capacitances (0.25 μm CMOS)

Bottom plate

Top plate

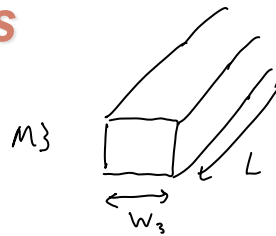
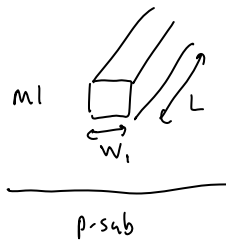
	Field	Active	Poly	Al1	Al2	Al3	Al4
Poly	88	$aF/\mu\text{m}^2$					
	54	$aF/\mu\text{m}$					
Al1	30	41	57				
	40	47	54				
Al2	13	15	17	36			
	25	27	29	45			
Al3	8.9	9.4	10	15	41		
	18	19	20	27	49		
Al4	6.5	6.8	7	8.9	15	35	
	14	15	15	18	27	45	
Al5	5.2	5.4	5.4	6.6	9.1	14	38
	12	12	12	14	19	27	52

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25

Wiring Capacitances



$$C = C_{pp} \cdot W \cdot L + 2C_{fringe} L$$

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26